The upper limit of artificial intelligence

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Grothendieck universe and $\infty-$ Category

Collection x and y have relationship,

$$\begin{cases} x \nsim y \\ x \sim y: \bigwedge\limits_{i\in I\in U} H_i \end{cases} or\ (x,y)_{(\nsim,\sim)}: \bigwedge_{i\in I\in U} H_i \quad x,y,\nsim,\sim,:,H_i \in Grothendieck\ universe$$

Grothendieck universe is a transitive set, and $\emptyset \simeq \nsim$.

1 $x: y, y: z \Longrightarrow x: z$

2 $x, y \in U \Longrightarrow \{x, y\} \in U$

3 $x \in U \Longrightarrow x^* \in U$

4 $f: U \ni I \longrightarrow U \qquad \cup f(I) \in U$

$\{x, y\},\ \{\nsim, \sim\}$ satisfy

$$\{x, y\} \rightleftarrows \{\nsim, \sim\}\ or\ \{x, y\} \circ \{\nsim, \sim\}$$

$$(\{x, y\}: \{\nsim, \sim\}^*, \{\nsim, \sim\}: \{x, y\}^*)_{(\nsim,\sim)} \quad where\ (:)_{(\nsim,\sim)} \in U$$

The above formula reveals the hidden relationship between the whole and the parts.

$$m_n \simeq (m: n)\ or\ (n: m)\ \ where\ (m: n^*, n: m^*)_{(\nsim,\sim)}$$

The diagram is as follows,

$$\begin{array}{ccc} * & \longrightarrow & *^* \\ \uparrow\uparrow & & \downarrow\downarrow \\ *^* & \longleftarrow & * \end{array}$$

If $\nsim$ and $\sim$ are objects, then

$$(\{(\nsim), (\sim)\}: \{\nsim, \sim\}^*, \{\nsim, \sim\}: \{(\nsim), (\sim)\}^*)_{(\nsim,\sim)}$$

The left-associative relationship are as follows,

$$\{\nsim, \sim\}_{\because\{\nsim,\sim\}}: \mathcal{H} \longrightarrow \left(\{\nsim, \sim\}_{\because\{\nsim,\sim\}}\right)_{\mathcal{H}}$$

If $x \longrightarrow x \rightleftarrows * \longrightarrow (x, *)_{(\nsim,\sim)} \longrightarrow x_{(\nsim,\sim)} \longrightarrow x_{\nsim}, x_{\sim}$, then

$$x_{(\nsim,\sim)} \wedge \left(\prod_i \mathcal{C}_i\right)_{(\nsim,\sim)} \longrightarrow \left(\prod_i \{x, \mathcal{C}_i\}\right)_{(\nsim,\sim)} : \mathcal{M} \longrightarrow \mathcal{M}_{(\nsim,\sim)} \longrightarrow \cdots$$

$$where\ \prod * \simeq (*) \simeq +,\ \wedge * \simeq \{*\} \simeq \times$$

Exactly,

$$\nwarrow \overset{F}{\underset{G}{\rightleftarrows}} \emptyset \quad and \;\; Hom_{\emptyset}(F(C), D) \simeq Hom_{\nwarrow}\big(C, G(D)\big)$$

$$\begin{array}{ccccccccc} x & \longrightarrow & x_{\nwarrow} & \longrightarrow & (\{x, \mathcal{C}\})_{\nwarrow} & \simeq & (x_{\mathcal{C}})_{\nwarrow} & \simeq & x_{(\mathcal{C}_{\nwarrow})} \\ & & & \nwarrow & & & & & \downarrow \\ & & & & x_{\nwarrow} & \longleftarrow & x_{\emptyset} & \longleftarrow & x_{(\mathcal{C}_{\emptyset})} \end{array}$$

One has

$\nwarrow_{\nwarrow}, \sim_{\nwarrow}, \nwarrow_{\sim}, \sim_{\sim} : U \Longrightarrow (\nwarrow_{\nwarrow})_{\nwarrow}, (\nwarrow_{\nwarrow})_{\sim}, (\sim_{\nwarrow})_{\nwarrow}, (\sim_{\nwarrow})_{\sim}, (\nwarrow_{\sim})_{\nwarrow}, (\nwarrow_{\sim})_{\sim}, (\sim_{\sim})_{\nwarrow}, (\sim_{\sim})_{\sim} : U$

The diagram is as follows,

$$\begin{array}{ccc} x & & \\ \downarrow & \searrow & \\ \{x, \nwarrow\} & \rightarrow & \big\{\{x, \mathcal{C}\}, \nwarrow\big\} \end{array} \Longrightarrow x_{(\nwarrow, \sim)} : \{x, \mathcal{C}\}$$

Repeat the above process,

$$(x_{\mathcal{C}})_{\nwarrow} : \mathcal{M} \longrightarrow \mathcal{M}_{\nwarrow}$$

$$\mathcal{C}_{\nwarrow} \longrightarrow (\{\mathcal{C}, \mathcal{D}\})_{\nwarrow} \simeq (\mathcal{C}_{\mathcal{D}})_{\nwarrow} \simeq \mathcal{C}_{(\mathcal{D}_{\nwarrow})} \simeq \big((\mathcal{C} \times \mathcal{D}) \times \nwarrow\big) \simeq \big(\mathcal{C} \times (\mathcal{D} \times \nwarrow)\big)$$

$$where \;\; \nwarrow \; is \; \infty - category$$

and

$$\mathcal{C}_{\nwarrow} \longrightarrow \left(\mathcal{C}_{\left(\mathcal{D}_1 \vdots \mathcal{D}_i\right)}\right)_{\nwarrow} \qquad \bigcup_i f(i) \in U \quad where \;\; f : I \ni i \longrightarrow \big(\mathcal{C} : (\mathcal{D}_1 : \cdots : \mathcal{D}_i)\big)_{\nwarrow} \in U$$

Transitivity and Interaction

If $\sim$, $\nwarrow \in U; \alpha, \beta \in Topology\, Y \Longrightarrow \sim_{\alpha}, \sim_{\beta} \in U$, then

$$(\sim) \nwarrow (\sim) \longrightarrow (\sim_{\alpha}) \nwarrow \big(\sim_{\beta}\big) \longrightarrow \big(\sim_{\alpha}, \sim_{\beta}\big)_{\nwarrow} \longrightarrow \big(\sim_{(\alpha,\beta)}\big)_{\nwarrow} \simeq \sim_{((\alpha,\beta)_{\nwarrow})}$$

1 $x \sim_{\alpha} y : L \;\; (x \longrightarrow L \longrightarrow y) \bigvee (y \longrightarrow L \longrightarrow x)$. If you remove either x or y, and L still exists, which has transitivity.

2 $x \sim_{\beta} y : P$ collection $x, y \in U \quad x \wedge y \longrightarrow P$

If there are no x, y such that P doesn't exist, then P is 'interaction'.
The four axioms of the grothendieck universe are roughly like collection $\big\{\sim_{\alpha}, \sim_{\beta}, * , f\big\}$, which is self-referential. The method for solving self-reference will end up getting caught in self-reference again. The limit of AI is this four axioms.

$$\big\{\sim_{\alpha}, \sim_{\beta}, * , f\big\}_{\nwarrow} \rightleftarrows \{\sim_{\alpha}, \nwarrow , * , f\}_{\sim_{\beta}} \longrightarrow \big\{\sim_{\alpha}, \sim_{\beta}\big\}_{\nwarrow} \rightleftarrows \{\sim_{\alpha}, \nwarrow\}_{\sim_{\beta}} \;\; where \;\; \nwarrow \rightleftarrows \sim_{\beta}$$

$$\sim_{\alpha} \wedge \sim_{\beta} \wedge * \wedge f : \big\{\sim_{\alpha}, \sim_{\beta}, * , f\big\} \longrightarrow \big\{\sim_{\alpha}, \sim_{\beta}, * , f\big\}^{*} : \sim_{\alpha^*} \wedge \sim_{\beta^*} \wedge *^{*} \wedge f^{*}$$

The diagram is as follows,

$$\begin{array}{ccc} \{\sim_\alpha, \sim_\beta, *, f\}_{\prec} & \longrightarrow & \{\sim_\alpha, \sim_\beta, *, f\}_{\prec}{}^{*} \\ \downarrow & & \downarrow \\ \{\sim_\alpha, \prec, *, f\}_{\sim_\beta} & \longrightarrow & \{\sim_\alpha, \prec, *, f\}_{\sim_\beta}{}^{*} \end{array}$$

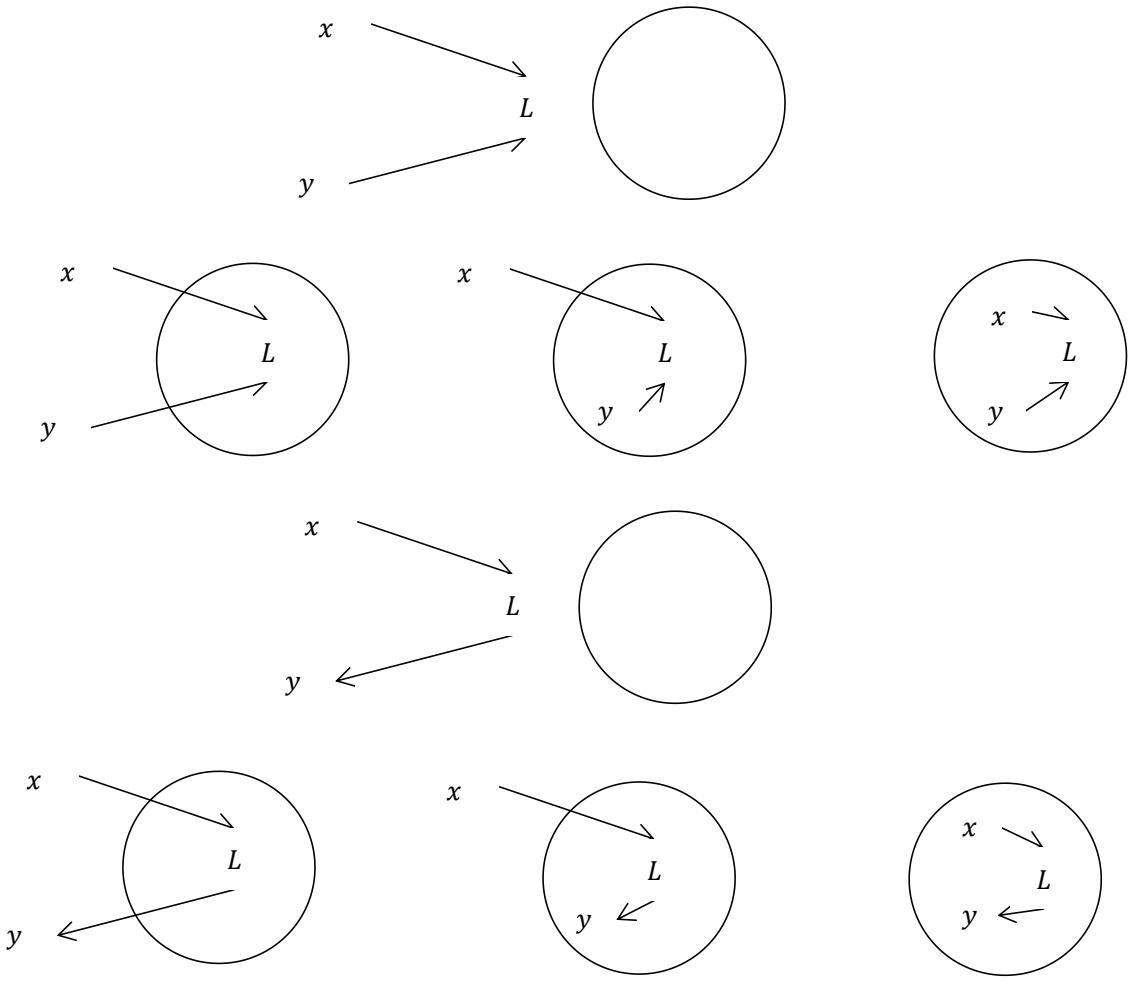
Other examples like $(\sim_\alpha: (\sim_\beta)^{*}, \sim_\beta: (\sim_\alpha)^{*})_{(\prec,\sim)}$, $(\sim_\alpha: *^{*}, * : (\sim_\alpha)^{*})_{(\prec,\sim)}$ are not listed one by one. Concepts come from differrences, so collection

$$x, c: \{x, \{\prec, c\}\}: \{\prec, c\} \; or \; x: \bigwedge_c (x, c)_{\prec} : \bigwedge_c (*, c)_{\prec}$$

$$\mathcal{F}: I \ni c \longrightarrow c^{\prec} := (*, c)_{\prec} \in U \Longrightarrow x: \bigcup_c \mathcal{F}(c) : U$$

The nonexistence of a closed system and the irreducibility of emergence

There are eight kinds of relationships between objects and systems.



Let's consider a simple scenario.The same bouncy ball Q falls on the same carpet $\mathcal{A}$, but there are different things b_1 and b_2 hidden under the carpet, and the system doesn't know about them.

$$\begin{cases} Q_1 \sim_\alpha Q_2 : Q \\ a_1 \sim_\alpha a_2 : a : \mathcal{A} \end{cases}$$

$$\begin{cases} Q_1 \sim_\beta a_1 : \{Q_1, a_1\} \\ Q_2 \sim_\beta a_2 : \{Q_2, a_2\} \end{cases} \sim_\alpha \begin{cases} e_1 \sim_\beta \{Q_1, a_1\} \\ e_2 \sim_\beta \{Q_2, a_2\} \end{cases} \sim_\alpha \begin{cases} h_1 \sim_\beta a_1 : \{h_1, a_1\} \\ h_2 \sim_\beta a_2 : \{h_2, a_2\} \end{cases} \sim_\alpha \begin{cases} e_1 \sim_\beta b_1 \\ e_2 \sim_\beta b_2 \end{cases}$$

$$Q \sim_\beta a : \Lambda_{i \in I} \{Q, a\}_i$$

$$\forall i, \{Q, a\}_i : (Q, a)_{\sim_\beta} \longrightarrow (\cdots, \{Q, a\}_i, \cdots)_\nu = \left(\prod_i \{Q, a\}_i\right)_\nu : \left((Q, a)_{\sim_\beta}\right)_\nu$$

The system finds new relationships from the results of differences, like the intertwining of transitivity and interaction, which can't be rerified. The point where axioms appear is exactly where the outside world intervenes in the system.

$$\begin{array}{ccc} Q & & \\ \downarrow & \searrow & \\ Q \sim_\beta a & \rightarrow & Q \sim_\beta \{a, \mathcal{B}\} \end{array}$$

$$\left(\left((Q, a)_{\sim_\beta}\right)_\nu, (b_1, b_2)_\nu\right)_{\sim_\alpha} \longrightarrow \left(Q, \left(a, (b_1, b_2)_{\sim_\alpha}\right)_{\sim_\beta}\right)_{\sim_\beta}$$

$$\left(\nu_{\sim_\alpha} \longrightarrow \nu_{\sim_\beta}\right) \simeq \left(\nu_\sim \longrightarrow \nu_\nu\right)$$

As long as it's defined, $\sim_\beta$ falls into $\sim_\alpha$. The diagram is as follows,

$$\begin{array}{ccc} \left(\left((*)_{\sim_\beta}\right)_*, *\right)_* & \rightarrow & \left(*, *_{\sim_\beta}\right)_{\sim_\beta} \\ \downarrow & & \downarrow \\ \left(\left((*)_{\sim_\alpha}\right)_*, *\right)_* & \rightarrow & \left(*, *_{\sim_\alpha}\right)_{\sim_\alpha} \end{array} \qquad where \{\sim_\alpha, \sim_\beta, *, f\} : *$$

When $\sim_\beta$ changes position, axiom cannot prove itself. Denying the law of identity would lead to the emergence of infinite categories, which can't exist in a closed system. This reveals the contradiction between static definitions and dynamic changes.

$$\left(\sim_\beta\right)_* \longrightarrow (*)_{\sim_\beta}$$

Reference

1. Gödel, Kurt. "Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I." Monatshefte für mathematik und physik 38.1 (1931): 173-198.
2. Lurie, Jacob. Higher topos theory. Princeton University Press, 2009.